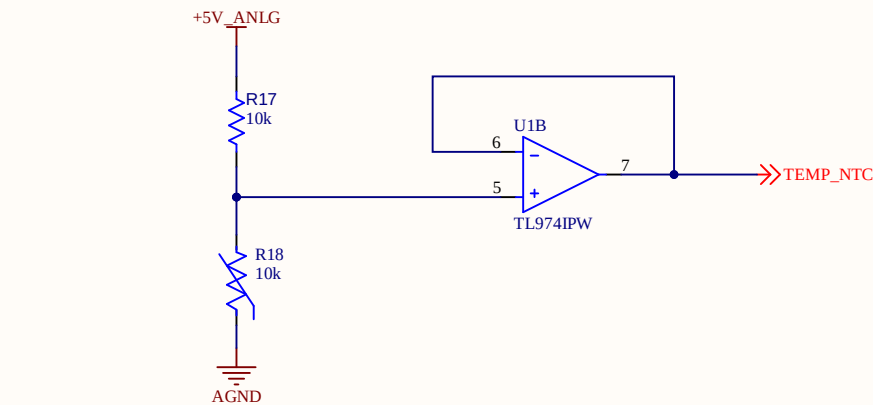
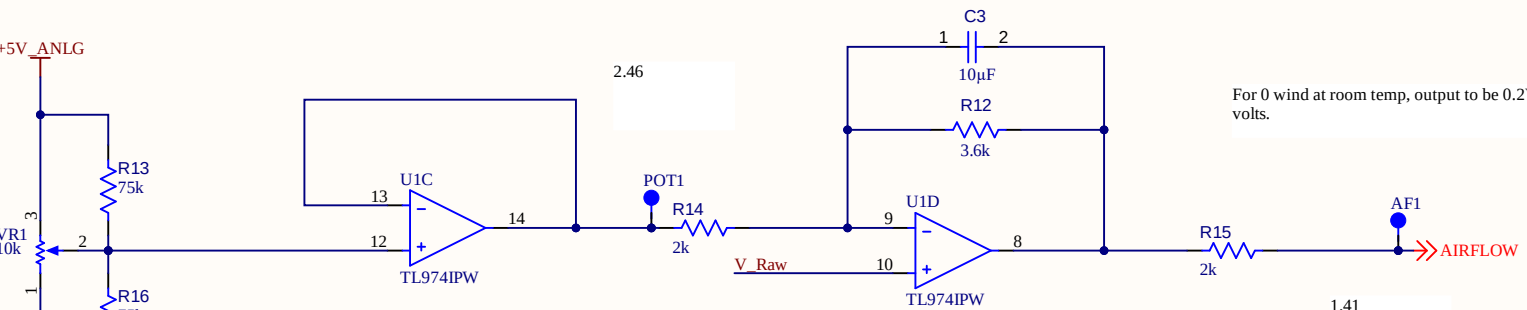
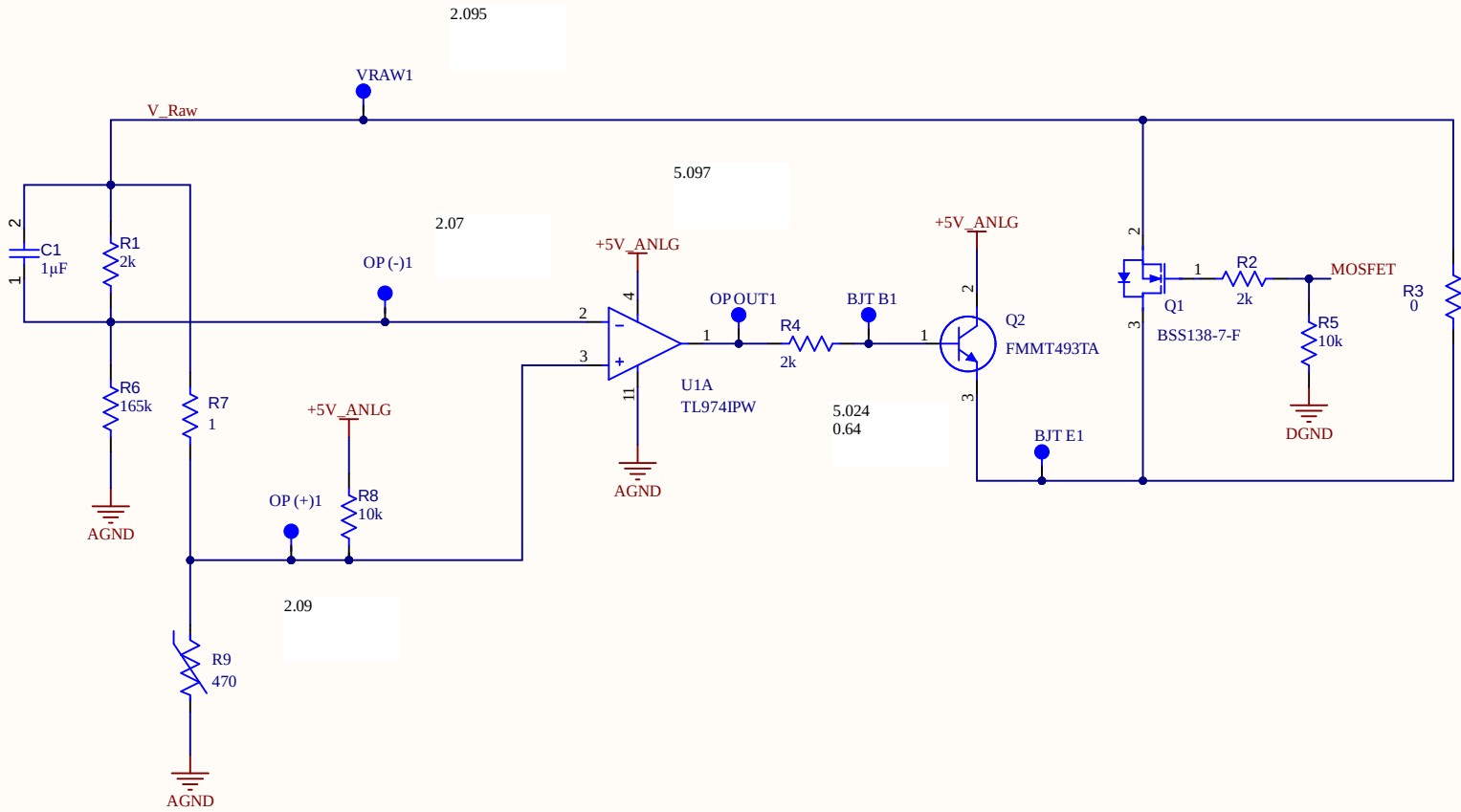
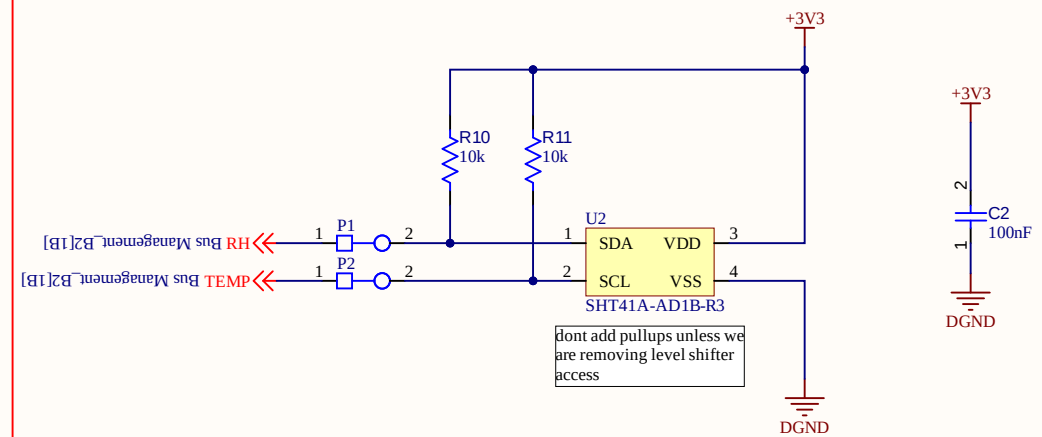


# Airflow Sensor Circuit



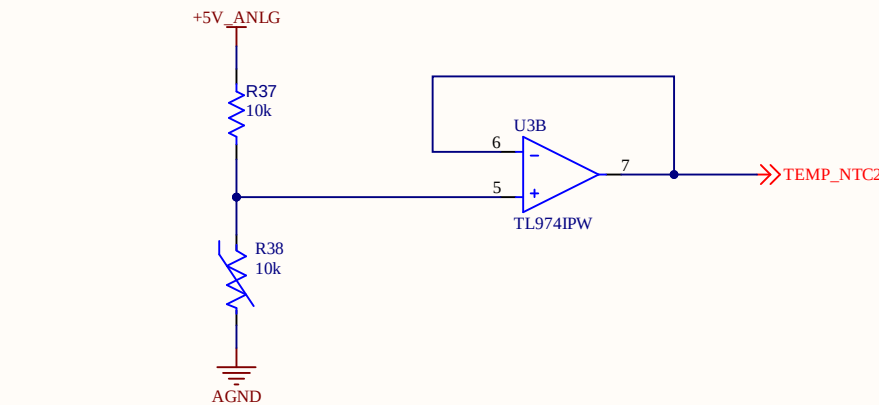
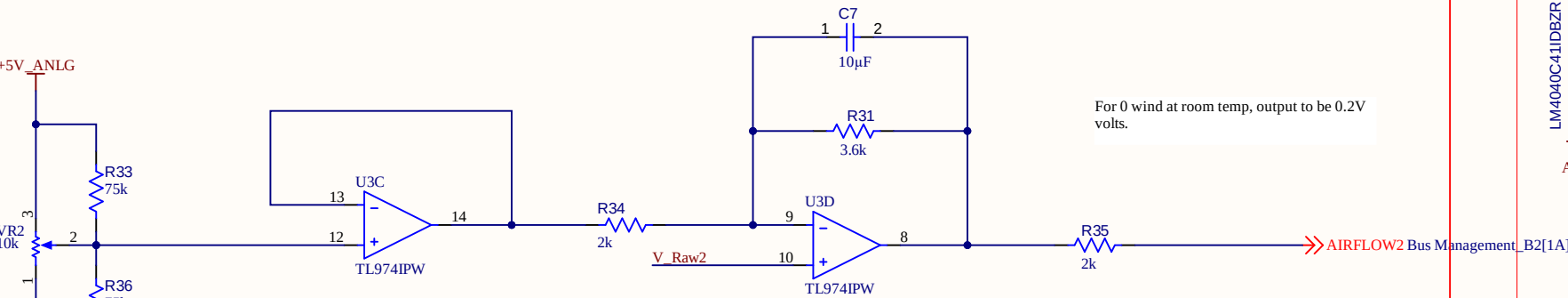
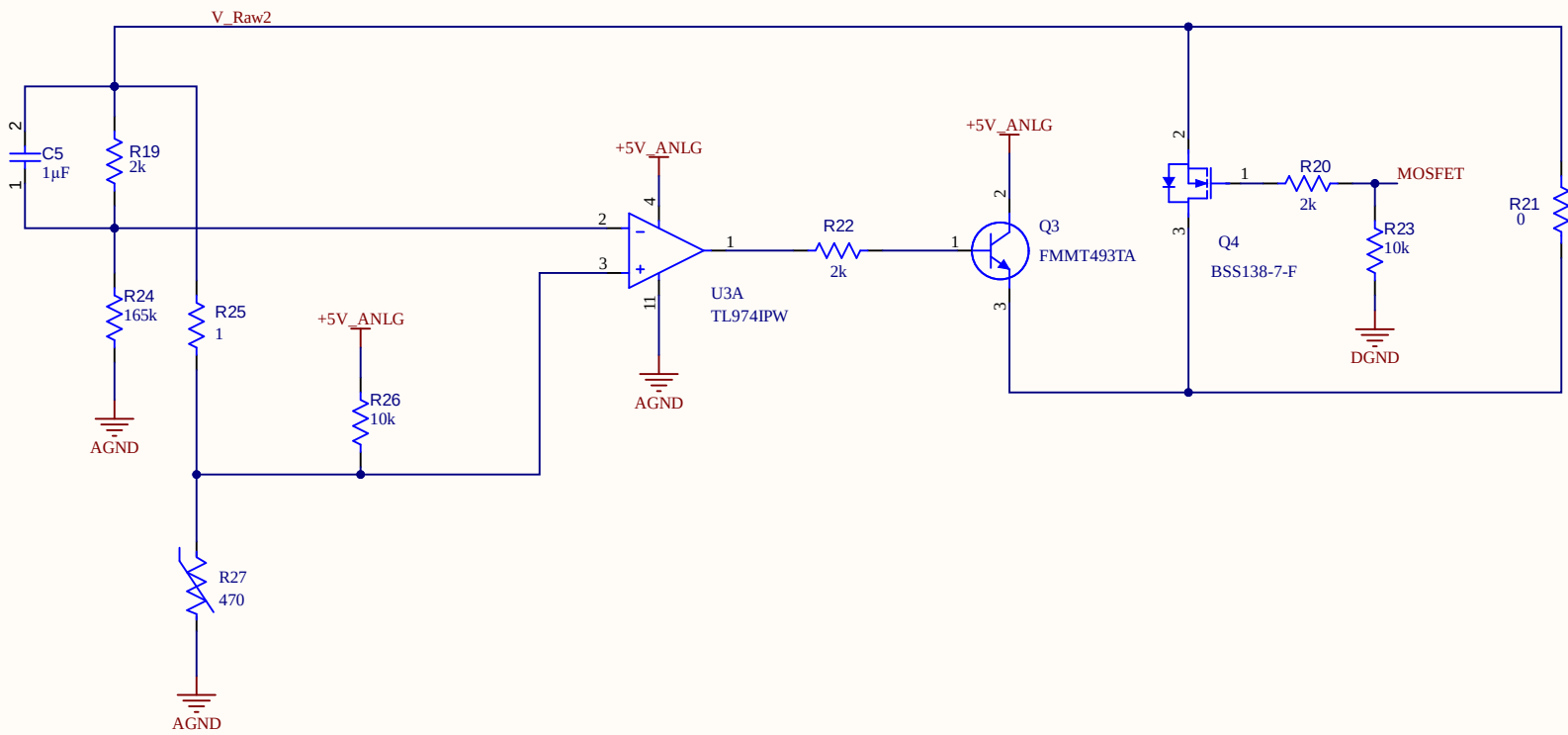
# Temp/Humidity Sensor



16 bit  
Dont forget cutout region  
0x44 address  
SCL - Temp  
SDA - RH  
100-400Khz

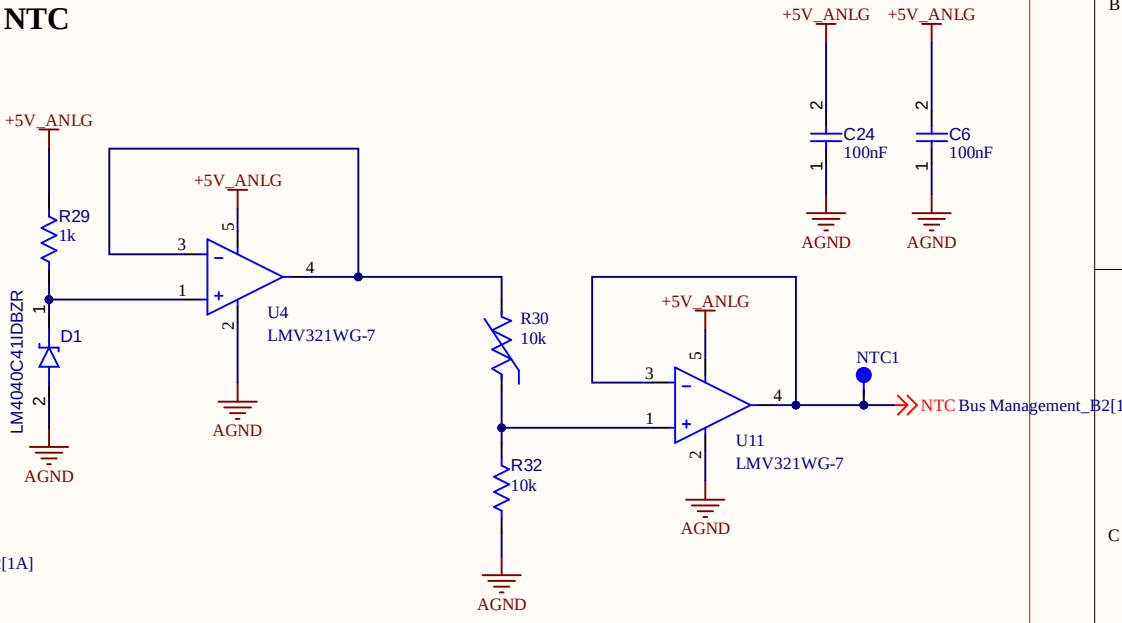
|                                         |                 |             |
|-----------------------------------------|-----------------|-------------|
| Title SAMBA 2 remote board - Airspeed 1 |                 |             |
| Size A3                                 | Number          | Revision B1 |
| Date: 8-21-2024                         | Sheet of 2/3    |             |
| File: C:\Users\...\Airspeed1_B2.SchDoc  | Drawn By: K.A.I |             |

Airflow Sensor Circuit Back



For 0 wind at room temp, output to be 0.2V volts.

NTC



|                                            |                 |                |
|--------------------------------------------|-----------------|----------------|
| Title<br>SAMBA 2 remote board - Airspeed 2 |                 |                |
| Size<br>A3                                 | Number          | Revision<br>B1 |
| Date:<br>8-21-2024                         | Sheet of 3/3    |                |
| File:<br>C:\Users\...\Airspeed2_B2.SchDoc  | Drawn By: K.A.I |                |

### ADC for airflow sensor sigs

ADD is 1001011

B sides can be connected together  
 $V_{ca} < V_{cb} - 1$  - here  $v_{ca}$  is 3.3V  
 The A side doesn't need pull ups, but the B side does

### Level shifter

Target side

Micro side

B sides can be connected together  
 $V_{ca} < V_{cb} - 1$  - here  $v_{ca}$  is 3.3V  
 The A side doesn't need pull ups, but the B side does

[illegible]

### 3.3V Regulator and Power LED

The diagram illustrates a 3.3V regulator and power LED circuit. The regulator, U10 (LD39200PU33R), is powered by a +5V source through a 10µF capacitor (C19). Its output provides +3V3 to a power LED (D7, AA3528CGSK) through a 2k resistor (R49). The LED's cathode is grounded (DGND). A note indicates the polarity notch on the LED is towards the K terminal.

## 5V Filter and Ground Connection

The left diagram shows five pins connected to ground symbols:

- +5V
- +3V3
- +5V\_ANLG
- AG
- DG

The right diagram shows a detailed 5V filter circuit:

- Input: +5V
- Resistor: R50 (0.91)
- Inductor: L1 (2.2uH)
- Capacitors: C21 (10uF), C22 (1uF), C23 (100nF) in parallel
- Output: +5V\_ANLG
- Ground: AGND

|                                                    |                                       |                 |
|----------------------------------------------------|---------------------------------------|-----------------|
| Title <b>SAMBA 2 remote board - Bus Management</b> |                                       |                 |
| Size<br>A3                                         | Number                                | Revision<br>B1  |
| Date:                                              | 8-21-2024                             | Sheet of 1/3    |
| File:                                              | C:\Users\...\Bus Management_B2.SchDoc | Drawn By: K.A.I |